

5.1 HW: Maclaurin Series Take-home — Markscheme

4. [Maximum mark: 21]

AA HL 2022 May P1 TZ1 Q12

(a) Maclaurin series for $f(x) = e^x \sin x$ up to x^3 . [4]

Method 1 (multiply known series). M1

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots, \quad \sin x = x - \frac{x^3}{3!} + \cdots$$

Multiply, keeping terms up to x^3 : M1

$$\begin{aligned} e^x \sin x &= \left(1 + x + \frac{x^2}{2} + \frac{x^3}{6}\right)\left(x - \frac{x^3}{6}\right) + \cdots \\ &= x + x^2 + \frac{x^3}{2} - \frac{x^3}{6} + \cdots \text{A1} \\ &= x + x^2 + \frac{1}{3}x^3 + \cdots \text{A1} \end{aligned}$$

Method 2 (derivatives at 0). Use $f^{(n)}(0)$ to build the series: $f(0) = 0$, $f'(0) = 1$, $f''(0) = 2$, $f'''(0) = 2$, giving the same answer.

(b) Approximate $\int_0^1 e^{x^2} \sin(x^2) dx$. [4]

Solution. Substituting $x \rightarrow x^2$ in (a):

$$e^{x^2} \sin(x^2) \approx x^2 + x^4 + \frac{1}{3}x^6. \text{A1}$$

Integrate termwise: M1

$$\int_0^1 (x^2 + x^4 + \frac{1}{3}x^6) dx = \left[\frac{x^3}{3} + \frac{x^5}{5} + \frac{x^7}{21}\right]_0^1 = \frac{1}{3} + \frac{1}{5} + \frac{1}{21} \text{A1}$$

Common denominator 105: $\frac{35 + 21 + 5}{105} = \frac{61}{105}$. A1

(c)(i) Show $g''(x) = 2(g'(x) - g(x))$ where $g(x) = e^x \cos x$. [4]

Solution. Product rule: M1

$$g'(x) = e^x \cos x - e^x \sin x = e^x(\cos x - \sin x). \text{A1}$$

$$g''(x) = e^x(\cos x - \sin x) + e^x(-\sin x - \cos x) = -2e^x \sin x. \text{A1}$$

Then

$$2(g'(x) - g(x)) = 2[e^x(\cos x - \sin x) - e^x \cos x] = -2e^x \sin x = g''(x). \text{A1 AG}$$

(c)(ii) Deduce $g^{(4)}(x) = 2(g'''(x) - g''(x))$. [1]

Differentiating both sides of $g''(x) = 2(g'(x) - g(x))$ once gives $g'''(x) = 2(g''(x) - g'(x))$, and differentiating once more gives $g^{(4)}(x) = 2(g'''(x) - g''(x))$. **A1 AG**

(d) Maclaurin series for $g(x)$ up to x^4 . [5]

Solution. From (c):

- $g(0) = e^0 \cos 0 = 1$
- $g'(0) = e^0(\cos 0 - \sin 0) = 1$
- From $g''(x) = 2(g'(x) - g(x))$: $g''(0) = 2(1 - 1) = 0$ **A1**
- Differentiating: $g'''(x) = 2(g''(x) - g'(x))$, so $g'''(0) = 2(0 - 1) = -2$ **M1**
- And $g^{(4)}(0) = 2(g'''(0) - g''(0)) = 2(-2 - 0) = -4$ **A1**

Apply the Maclaurin formula: **M1**

$$g(x) = 1 + x + 0 \cdot \frac{x^2}{2} - \frac{2}{6}x^3 - \frac{4}{24}x^4 + \dots = 1 + x - \frac{1}{3}x^3 - \frac{1}{6}x^4 + \dots \text{A1}$$

(e) Evaluate $\lim_{x \rightarrow 0} \frac{e^x \cos x - 1 - x}{x^3}$. [3]

Solution. Using the series from (d): **M1**

$$\begin{aligned} \frac{e^x \cos x - 1 - x}{x^3} &= \frac{\left(1 + x - \frac{1}{3}x^3 - \frac{1}{6}x^4 + \dots\right) - 1 - x}{x^3} = \frac{-\frac{1}{3}x^3 - \frac{1}{6}x^4 + \dots}{x^3} \text{A1} \\ &= -\frac{1}{3} - \frac{1}{6}x + \dots \rightarrow -\frac{1}{3} \text{ as } x \rightarrow 0. \text{A1} \end{aligned}$$

5. [Maximum mark: 15]

AA HL 2025 Nov P3 TZ3 Q02 (a)–(e)

(a) Sum to infinity of $1 - x^2 + x^4 - x^6 + \dots$ for $|x| < 1$. [2]

Solution. Geometric series with first term $u_1 = 1$ and common ratio $r = -x^2$: **M1**

$$S_\infty = \frac{u_1}{1 - r} = \frac{1}{1 + x^2} \text{A1}$$

(b) Show that $\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots$. [3]

Solution. Integrate both sides of $\frac{1}{1 + x^2} = 1 - x^2 + x^4 - x^6 + \dots$:

$$\int \frac{1}{1 + x^2} dx = \arctan x + C, \text{A1}$$

$$\int (1 - x^2 + x^4 - x^6 + \dots) dx = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots + C'. \mathbf{A1}$$

At $x = 0$: $\arctan 0 = 0$, so $C = C' = 0$. Therefore $\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots$.

R1 AG

(c) Approximate π to three decimal places using $x = \frac{1}{\sqrt{3}}$ and the first three non-zero terms. [3]

Solution. Since $\tan \frac{\pi}{6} = \frac{1}{\sqrt{3}}$, $\arctan \frac{1}{\sqrt{3}} = \frac{\pi}{6}$. **A1**

$$\frac{\pi}{6} \approx \frac{1}{\sqrt{3}} - \frac{1}{3} \left(\frac{1}{\sqrt{3}} \right)^3 + \frac{1}{5} \left(\frac{1}{\sqrt{3}} \right)^5 = \frac{41}{45\sqrt{3}}. \mathbf{M1}$$

(GDC: evaluate the sum directly.) Multiplying by 6:

$$\pi \approx \frac{246}{45\sqrt{3}} = \frac{82\sqrt{3}}{45} \approx 3.156. \mathbf{A1}$$

(d) Number of terms needed so that the error in approximating $\arctan(1/\sqrt{3})$ is less than 0.0001. [3]

Solution. The m th non-zero term of the arctan series is $\frac{(-1)^{m-1}x^{2m-1}}{2m-1}$. By the alternating-series theorem, the error after m non-zero terms is at most

$$\left| \frac{x^{2m+1}}{2m+1} \right|. \mathbf{M1}$$

At $x = \frac{1}{\sqrt{3}}$, this becomes $\frac{1}{(2m+1)3^{(2m+1)/2}}$, and we need this to be less than 10^{-4} .

Tabulate on GDC:

m	error bound at $x = 1/\sqrt{3}$	
4	7.92×10^{-4}	M1
5	2.16×10^{-4}	
6	$6.09 \times 10^{-5} < 10^{-4}$	

So $m = 6$ non-zero terms are required.

A1

(e) By integration by parts, show

$$\int_0^{1/\sqrt{3}} \arctan x dx = \frac{\pi}{6\sqrt{3}} - \frac{1}{2} \ln \frac{4}{3}. [4]$$

Solution. Let $u = \arctan x$, $\frac{dv}{dx} = 1$, so $\frac{du}{dx} = \frac{1}{1+x^2}$ and $v = x$.

A1

Then

$$\int_0^{1/\sqrt{3}} \arctan x \, dx = [x \arctan x]_0^{1/\sqrt{3}} - \int_0^{1/\sqrt{3}} \frac{x}{1+x^2} \, dx. \mathbf{A1}$$

For the remaining integral, substitute $w = 1 + x^2$, $dw = 2x \, dx$, or recognise $\frac{x}{1+x^2} = \frac{1}{2} \cdot \frac{2x}{1+x^2}$:

$$\int_0^{1/\sqrt{3}} \frac{x}{1+x^2} \, dx = \frac{1}{2} [\ln(1+x^2)]_0^{1/\sqrt{3}} = \frac{1}{2} \ln\left(1 + \frac{1}{3}\right) = \frac{1}{2} \ln \frac{4}{3}. \mathbf{A1}$$

Also $[x \arctan x]_0^{1/\sqrt{3}} = \frac{1}{\sqrt{3}} \cdot \frac{\pi}{6} = \frac{\pi}{6\sqrt{3}}$. Combining,

$$\int_0^{1/\sqrt{3}} \arctan x \, dx = \frac{\pi}{6\sqrt{3}} - \frac{1}{2} \ln \frac{4}{3}. \mathbf{A1 \ AG}$$